

Jinwoo Jeong

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Research Interests

My research focuses on building efficient **AI systems** for deep neural networks and **large language models (LLMs)**, particularly through **memory management** and **request scheduling** in GPU-centric environments.

◦ LLM Serving Systems ◦ System for Machine Learning ◦ Operating Systems ◦ Computer Architecture

Education

Korea University

Seoul, South Korea

Ph.D. Student in Electrical and Computer Engineering

Sep. 2024 – Feb. 2027

◦ Advisor: Prof. Jeongseob Ahn

◦ Transferred from Ajou University, Aug. 2024

Ajou University

Suwon, South Korea

Ph.D. Student in Computer Engineering

Mar. 2020 – Aug. 2024

Ajou University

Suwon, South Korea

B.E. in Software and Computer Engineering

Mar. 2016 – Feb. 2020

Publications

Jinwoo Jeong and Jeongseob Ahn. “An Efficient DNN Model Serving System using Layer-wise Caching and Direct-Host-Access”, *ACM Transactions on Computer Systems (TOCS)*, 2025.

Jinwoo Jeong and Jeongseob Ahn. “Accelerating LLM Serving for Multi-turn Dialogues with Efficient Resource Management”, *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2025.

Woohyung Choi, **Jinwoo Jeong**, Hanhwi Jang, and Jeongseob Ahn. “GPU-centric Memory Tiering for LLM Serving with NVIDIA Grace Hopper Superchip”, *IEEE Computer Architecture Letters (CAL)*, 2025.

Jinwoo Jeong, Seungsu Baek, and Jeongseob Ahn. “Fast and Efficient Model Serving Using Multi-GPUs with Direct-Host-Access”, *ACM European Conference on Computer Systems (EuroSys)*, 2023.

Sangjin Choi, Taeksoo Kim, **Jinwoo Jeong**, Rachata Ausavarungnirun, Myeongjae Jeon, Youngjin Kwon, and Jeongseob Ahn. “Memory Harvesting in Multi-GPU Systems with Hierarchical Unified Virtual Memory”, *USENIX Annual Technical Conference (ATC)*, 2022.

Preprints

Jinwoo Jeong, Woohyung Choi, Myeongjae Jeon, and Jeongseob Ahn. “Taming Speculative Search for Test-Time Scaling in LLM Serving”, arXiv:2609.39334, 2026 (*Under Review*).

Open-source Projects and Contributions

SGLang | <https://github.com/sgl-project/sglang>

◦ Contributed to an open-source LLM serving framework for efficient and scalable inference

LMCache | <https://github.com/LMCache/LMCache>

- Contributed to an open-source KV cache management system for LLM serving

LMServe | <https://github.com/Sys-KU/LMServe>

- Developed an open-source LLM serving framework that supports multi-level KV caching, request reordering, and disaggregated inference for efficient multi-turn LLM serving.

DeepPlan | <https://github.com/Sys-KU/DeepPlan>

- Released the EuroSys'23 artifact for multi-GPU DNN model serving, implementing layer-wise caching, direct-host-access, and parallel transmission to improve inference efficiency.

Awards

Best Paper Award Runner-up, <i>IEEE Computer Architecture Letters</i>	2025
Research Excellence Award, <i>Korea University</i>	2025
Gilles Muller Best Artifact Award, <i>EuroSys</i>	2023

Academic Services

Journal Reviewer

- 2026: IEEE Computer Architecture Letters

Shadow Program Committee

- 2026: EuroSys

Technical Skills

Languages: C/C++, Python, Rust, CUDA, Triton, Go

ML Frameworks: SGLang, vLLM, LMCache, PyTorch, FlashInfer, FlashAttention

Systems & Libraries: NVIDIA UVM Driver, Linux Kernel, NIXL, NCCL, Asyncio, gRPC, Boost

Developer Tools: CMake/Make, Git, Docker, Nsight Systems/Compute, GDB, Perf, PyTest